

MASTER CONTRACT DEVELOPMENT AND COMMERCIAL MANUFACTURING AGREEMENT (CDMO)

Agreement Number: PHARMA-CDMO-2028-MSA-4012

Effective Date: January 15, 2028

Initial Expiration Date: January 14, 2038

Renewal Option: Two optional three-year renewals upon 180 days written notice

Sponsor: BioVanguard Therapeutics Inc., a Delaware corporation ("Sponsor")

CDMO Provider: Apex Pharma Solutions LLC, a Delaware limited liability company ("CDMO")

Primary Manufacturing Facility: Apex Biomanufacturing Campus, Research Triangle Park, North Carolina

Contract Currency: United States Dollars (USD)

Governing Law: State of Delaware, excluding conflict of laws principles

Regulatory Compliance Standards: US FDA 21 CFR Parts 210/211 (cGMP), EMA EudraLex Volume 4, ICH Q7/Q8/Q9/Q10

RECITALS

WHEREAS, Sponsor is a biopharmaceutical company engaged in the clinical development and commercial distribution of novel monoclonal antibodies, recombinant proteins, and cell-based therapeutic products;

WHEREAS, CDMO possesses state-of-the-art biopharmaceutical process development laboratories, mammalian cell culture suites, aseptic fill-finish lines, quality control testing facilities, and cold-chain storage infrastructure operating under current Good Manufacturing Practices (cGMP);

WHEREAS, Sponsor desires to engage CDMO, and CDMO agrees to be engaged by Sponsor, to perform process transfer, analytical validation, commercial drug substance manufacturing, aseptic drug product filling, stability testing, and release testing under strict key performance indicators (KPIs) and quality agreements;

NOW, THEREFORE, in consideration of the mutual promises, covenants, and legal conditions set forth herein, the Parties agree as follows:

ARTICLE I: DEFINITIONS AND cGMP COMPLIANCE RULES

Section 1.01: Rules of Interpretation

(a) Terms defined in FDA 21 CFR Parts 210/211, ICH guidelines, or EudraLex Volume 4 shall carry their statutory cGMP meanings unless explicitly modified herein.

(b) All references to days mean calendar days unless specified as business days.

Section 1.02: Defined Terms

As used in this Agreement, the following defined terms shall have the specified meanings:

- 1. Acceptance Criteria:** Numerical limits, ranges, or other suitable measures for testing the safety, identity, strength, quality, and purity of Biopharmaceutical Drug Products.
- 2. Active Pharmaceutical Ingredient (API):** Any substance or mixture of substances intended to be used in the manufacture of a drug product that becomes an active ingredient of the drug product.
- 3. Batch Right-First-Time (RFT):** The percentage of completed commercial manufacturing lots that meet all batch release criteria and quality specifications without incurring critical deviations or requiring re-processing.
- 4. Bulk Drug Substance (BDS):** Unformulated active biopharmaceutical material produced by cell culture harvest and purification prior to final formulation and filling.
- 5. Cleanroom ISO Class 5 Uptime:** The percentage of operational hours during a billing month that critical filling suites (Grade A laminar flow hoods) maintain ISO 5 airborne particle count compliance ($< 3,520 \text{ particles/m}^3 \geq 0.5 \mu\text{m}$).
- 6. Critical Quality Attribute (CQA):** A physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure desired product quality.
- 7. Deviation Lead Time:** The number of calendar days elapsed between the initial logging of an out-of-trend event or process parameter drift and the formal approval of the investigation report by CDMO Quality Assurance.
- 8. Excursion Event:** Any temperature, humidity, or differential pressure reading outside validated operating ranges lasting longer than five (5) consecutive minutes in any stability chamber, cold storage room, or bioreactor suite.
- 9. Failure to Supply:** Any inability of CDMO to deliver at least ninety percent (90%) of a confirmed purchase order lot volume within ten (10) calendar days of the agreed Delivery Deadline.
- 10. Good Manufacturing Practice (cGMP):** Current Good Manufacturing Practices enforced by the FDA under 21 CFR Parts 210 and 211, and EMA EudraLex Volume 4.
- 11. Master Batch Record (MBR):** Document containing the master manufacturing instructions, bill of materials, sampling plans, and quality controls for a commercial lot.
- 12. On-Time In-Full (OTIF) Delivery:** The percentage of commercial drug product lots delivered to Sponsor's designated distribution hubs on or before the agreed delivery date with all required certificate of analysis (CoA) documentation.
- 13. Out-of-Specification (OOS) Result:** Any analytical test result that fails to meet pre-established specification limits documented in approved regulatory filings.
- 14. Process Analytical Technology (PAT):** Systems for designing, analyzing, and controlling manufacturing through timely measurements of critical quality parameters.
- 15. Process Yield Efficiency:** The percentage ratio of actual purified protein yield achieved in a commercial bioreactor run to the theoretical maximum yield validated during process characterization.

- 16. Quality Agreement:** The detailed regulatory agreement executed concurrently herewith defining specific cGMP responsibilities between Sponsor and CDMO.
- 17. Stability Testing Chamber:** Temperature and humidity controlled environmental chambers maintained under ICH Q1A(R2) conditions for long-term real-time stability monitoring.
- 18. Master Cell Bank (MCB):** Preserved biological cell repository used as the starting material for all commercial fermentation runs.
- 19. Validation Master Plan (VMP):** Summary document detailing the validation strategy, cleaning validation, process performance qualification (PPQ), and facility qualification.
- 20. Certificate of Analysis (CoA):** Formal document signed by CDMO Quality Assurance certifying that a specific batch meets all tested specifications.

ARTICLE II: SCOPE OF BIOMANUFACTURING SERVICES

Section 2.01: Commercial Manufacturing Operations

CDMO shall manufacture, test, store, and package Sponsor's proprietary biologic drug substance and sterile injectable drug product at the RTP Campus. Services include:

- (a) Operation of dual 2,000-Liter single-use bioreactor trains for cell culture harvest;
- (b) Downstream purification including Protein A affinity chromatography, viral inactivation, tangential flow filtration (TFF), and bulk formulation;
- (c) Aseptic vial fill-finish operating at speeds up to 120 vials per minute under ISO 5 conditions;
- (d) Quality control release testing, sterility testing, mycoplasma testing, viral clearance testing, and stability storage.

ARTICLE III: COMMERCIAL TERMS AND BATCH PRICING

Section 3.01: Batch Fee Schedule

Production Category	Batch Batch Size	Fixed Commercial Batch Price	Billing Schedule
Bulk Drug Substance (BDS) Batch	2,000-Liter Bioreactor Run	\$425,000.00	30% at order, 70% at release
Aseptic Fill-Finish Lot	15,000 Vials (10mL)	\$185,000.00	50% at fill, 50% at CoA release
Stability Testing Storage Suite	Per ICH Condition / Year	\$14,500.00	Annual advance billing
Method Validation & Tech Transfer	Lump Sum Protocol	\$250,000.00	Milestone delivery payments

ARTICLE IV: KEY PERFORMANCE INDICATORS AND QUALITY METRICS

Section 4.01: KPI Performance Table

KPI ID	Metric Name	Target Value	Minimum Acceptable	Calculation Window
KPI-CDM-01	Batch Right-First-Time (RFT) Rate	>= 98.50%	< 95.00%	Quarterly Aggregate
KPI-CDM-02	On-Time In-Full (OTIF) Delivery	>= 99.00%	< 96.00%	Monthly Aggregate
KPI-CDM-03	Bioprocess Cell Yield Efficiency	>= 94.50%	< 90.00%	Per Batch Bioreactor Run
KPI-CDM-04	Cleanroom ISO Class 5 Uptime	99.99%	< 99.90%	Monthly Aggregate
KPI-CDM-05	Stability Excursion Drift Rate	0.00%	> 0.01%	Continuous Monitoring
KPI-CDM-06	OOS Investigation Lead Time	<= 10.0 Days	> 15.0 Days	Per Event
KPI-CDM-07	Critical Deviation Closure Lead Time	<= 14.0 Days	> 21.0 Days	Per Event
KPI-CDM-08	FDA/EMA Inspection Findings	0 Critical	> 0 Critical	Per Regulatory Audit
KPI-CDM-09	Product Complaint Rate	< 0.05%	> 0.10%	Distributed Dosage Units
KPI-CDM-10	Analytical Testing Release Lead Time	<= 7.0 Days	> 12.0 Days	Per Batch Sampling

ARTICLE 5: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 1

Section 5.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 5.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 5.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 5.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 5.05: Subcontractor Oversight and Vendor Qualification Rules

CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control, non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 5.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 6: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 2

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ARTICLE 7: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 3

Section 7.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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Section 7.02: Preventive Maintenance and Quality Control Verification

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ARTICLE 8: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 4

Section 8.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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Section 8.02: Preventive Maintenance and Quality Control Verification

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ARTICLE 9: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 5

Section 9.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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Section 9.02: Preventive Maintenance and Quality Control Verification

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ARTICLE 10: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 6

Section 10.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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ARTICLE 11: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 7

Section 11.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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ARTICLE 12: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 8

Section 12.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

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ARTICLE 13: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 9

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ARTICLE 14: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 10

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ARTICLE 15: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 11

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Section 15.02: Preventive Maintenance and Quality Control Verification

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Section 15.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 15.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 15.05: Subcontractor Oversight and Vendor Qualification Rules

CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control,

non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 15.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 16: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 12

Section 16.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 16.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

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Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 16.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

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CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control, non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 16.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 17: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 13

Section 17.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 17.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 17.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 17.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 17.05: Subcontractor Oversight and Vendor Qualification Rules

CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control, non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 17.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 18: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 14

Section 18.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 18.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 18.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 18.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 18.05: Subcontractor Oversight and Vendor Qualification Rules

CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control, non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 18.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 19: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 15

Section 19.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 19.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 19.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 19.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 19.05: Subcontractor Oversight and Vendor Qualification Rules

CDMO shall not subcontract, delegate, or assign any primary operational duties under this Agreement without Sponsor's express prior written approval. All approved subcontractors must be bound by written agreements enforcing quality control, non-disclosure, cybersecurity, and insurance standards at least as stringent as those set forth herein. CDMO remains fully liable for all subcontractor acts and omissions.

Section 19.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition

documentation to Sponsor annually.

ARTICLE 20: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 16

Section 20.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 20.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 20.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

Sponsor and its authorized independent auditors shall have unrestricted rights to inspect, audit, test, and evaluate CDMO's facilities, records, telemetry streams, and operating procedures upon five (5) business days written notice. In the event of an unannounced inspection by a federal or state regulatory authority (e.g., FDA, FCC, SEC, EPA, OSHA), CDMO shall notify Sponsor within two (2) hours of inspector arrival.

Section 20.04: Incident Escalation, Root Cause Analysis (RCA), and Corrective Action Plans

In the event of any operational breakdown, performance failure, security incident, or SLA breach, CDMO shall initiate immediate emergency response protocols. CDMO shall provide a preliminary incident notification within fifteen (15) minutes, contain the issue within thirty (30) minutes, and submit a comprehensive Root Cause Analysis (RCA) report within forty-eight (48) hours detailing permanent Corrective and Preventive Actions (CAPA).

Section 20.05: Subcontractor Oversight and Vendor Qualification Rules

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Section 20.06: Environmental Sustainability, Energy Efficiency, and Waste Disposal

CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

ARTICLE 21: OPERATIONAL GOVERNANCE, AUDIT AND COMPLIANCE - SECTION 17

Section 21.01: Detailed Operational Protocols in Biopharmaceutical CDMO Manufacturing

CDMO shall operate and manage all infrastructure, equipment, processes, software tools, and personnel engaged in Biopharmaceutical CDMO Manufacturing in accordance with established industry standards, strict safety protocols, and statutory requirements. All operational workflows shall be documented in standard operating procedures (SOPs) approved in writing by Sponsor prior to commercial execution.

Section 21.02: Preventive Maintenance and Quality Control Verification

CDMO shall maintain a comprehensive preventive maintenance and quality control regime. Equipment calibrations, diagnostic sweeps, software patches, and environmental audits shall occur on a mandatory recurring schedule. Detailed maintenance logs shall be preserved in tamper-evident electronic format for at least ten (10) years and made available for Sponsor inspection upon request.

Section 21.03: Regulatory Inspections, Statutory Reporting, and Audit Rights

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CDMO shall execute all operational activities in compliance with environmental protection regulations. CDMO shall implement energy-efficient practices, minimize hazardous waste generation, and provide certified waste disposition documentation to Sponsor annually.

SCHEDULE A: MANUFACTURING FACILITY AND CLEANROOM ROSTER

- 1. **Bioreactor Suites:** Suite A & Suite B featuring 2 x 2,000L Thermo Fisher DynaDrive Single-Use Bioreactors.
- 2. **Purification Suites:** Cleanroom Grade C downstream purification room equipped with ÄKTA process chromatography skids.
- 3. **Aseptic Filling Line:** Line 1 Isolated Restricted Access Barrier System (RABS) vial filling line operating under ISO Class 5 / Grade A conditions.
- 4. **Stability Testing Suite:** 12 walk-in ICH stability chambers (25°C/60% RH, 30°C/65% RH, 40°C/75% RH, 5°C, -20°C, -80°C).

SCHEDULE B: KPI PERFORMANCE GUARANTEES AND PENALTY MATRIX

Metric Code	KPI Target Name	Target	Penalty Threshold	Monetary Penalty / Remedy
CDM-01	Batch Right-First-Time	>= 98.5%	< 95.0%	\$150,000 credit per rejected batch + free re-run
CDM-02	OTIF Delivery	>= 99.0%	< 96.0%	\$250,000 / week Failure to Supply penalty
CDM-03	Bioprocess Yield	>= 94.5%	< 90.0%	Reimbursement of spoiled raw material API
CDM-04	Cleanroom Uptime	99.99%	< 99.90%	\$10,000 per hour of unapproved suite downtime
CDM-05	Stability Excursion Drift	0.00%	> 0.00%	Full batch replacement value if stability compromised
CDM-06	OOS Investigation Time	<= 10 Days	> 15 Days	\$5,000 per day over lead time limit
CDM-08	Regulatory Audit Findings	0 Critical	> 0 Critical	\$500,000 penalty + immediate Step-In Right

SIGNATURE PAGE

IN WITNESS WHEREOF, the Parties hereto have executed this Master Contract Development and Commercial Manufacturing Agreement as of the Effective Date.

BIOVANGUARD THERAPEUTICS INC.

By: _____

Name: Dr. Aris Thorne

Title: Chief Executive Officer

Date: January 15, 2028

APEX PHARMA SOLUTIONS LLC

By: _____

Name: Victoria Sterling

Title: Executive Vice President, Global Operations

Date: January 15, 2028